BAM
Bundesanstalt für Materialforschung und -prüfung

CERTIFICATE

CERTIFIED REFERENCE MATERIAL

BAM-K001 / Bottle No.

Ethanol in Water

Certified Value

Characteristic	Concentration at 20 °C in g/l	Uncertainty *U** at 20 °C in g/l
Ethanol	1,0291	0,0010

* Uncertainty *U* is the expanded uncertainty (coverage factor *k* = 2, corresponding to a confidence level of approximately 95%) and was calculated in accordance with the Guide to the Expression of Uncertainty in Measurement (GUM) [1]

End of Validity

The validity of this certificate ends 6 months after shipment of the material.
Date of shipment:

Material Description

The material BAM-K001 is a solution of ethanol in water, produced by gravimetric single-bottle dosing in units of approximately 4 liters. The solution is contained in a brown glass bottle with a screw cap according to DIN 45, equipped with a Teflon septum. The starting material used was ethanol absolute (≥ 99.8%) and water of quality grade 2 (DIN ISO 3696).

Recommended Use

The reference material is intended for the periodic calibration of breath alcohol analyzers in accordance with DIN VDE 0405-4 [1] and for applications according to the international recommendation OIML-R126 [2]. In addition, it can be used for the validation of analytical methods for the determination of ethanol in water.

Handling

When used in Evidential Breath Analysers the requirements of the relevant regulations must be observed. The material must be brought to ambient temperature before aliquots are taken and homogenized by shaking. The homogeneity of the material is ensured if the withdrawn

aliquot is equal to or greater than 10 ml. After an aliquot has been taken, the certified value for the remaining solution in the bottle can no longer be guaranteed.

Transport and Storage

The material should be stored tightly sealed at room temperature. It is recommended to keep the bottle in its cardboard packaging during storage. Direct exposure to sunlight must be avoided. The shelf life is 6 months from the date of shipment, provided the bottle remains sealed.

Participating Laboratories

The production of the reference material, homogeneity and stability studies, as well as the uncertainty estimation, were carried out at BAM, Department 1.

Analytical Methods

The certification is based on the gravimetric production process of the material. To confirm the weighing values and to investigate homogeneity and stability, gas chromatographic measurements with flame ionization detection (GC-FID) were performed with BAM reference procedure "Gas chromatographic precision determination (GC-FID) of ethanol in water" [3].

Metrological Traceability

The certified value (ethanol concentration in g/l at 20 °C) was calculated on a gravimetric basis. The balances are calibrated annually by an external accredited calibration laboratory. A buoyancy correction was applied. The density of the ethanol solution required for the calculation, depending on the ethanol mass fraction, was determined according to [4]. The purity of the ethanol was determined by GC-FID and Karl Fischer titration. The uncertainty of the ethanol concentration value was calculated from the uncertainty budget of the production process and the uncertainty of the purity. The uncertainty estimation was carried out in accordance with the guidelines of the GUM [5].

Literature

A detailed technical report describing the analysis procedures and the treatment of the analytical data used to certify BAM-K001-K007 is available on request or can be downloaded from https://webshop.bam.de/

[1] DIN VDE 0405-4: Ermittlung der Atemalkoholkonzentration - Teil 4: Prüfung von beweissicheren Atemalkohol-Messgeräten mit Prüfgas, 2019-07

[2] https://www.oiml.org/en/publications/recommendations/en/files/pdf_r/r126-e21.pdf

[3] https://www.bam.de/_SharedDocs/EN/Downloads/Reference%20procedures/4_gc-_fid.html

[4] F. Spieweck, H. Bettin: "Methoden zur Bestimmung der Dichte von Festkörpern und Flüssigkeiten" PTB-W-46, Braunschweig, Oktober 1991, 2. Nachdruck Februar 1998

[5] JCGM (2008) Evaluation of measurement data - guide to the expression of uncertainty in measurement, 1st edn. JCGM 100:2008 http://www.bipm.org/utils/common/documents/jcgm/JCGM_100_2008_E.pdf

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Bundesanstalt für Materialforschung und -prüfung (BAM)

Dr. S. Richter
Committee for Certification

Dr. R. Schneider
Project Coordinator
Division 1.8

This certificate is consistent with the capabilities that are included in Appendix C of the MRA drawn up by the CIPM. Under the MRA, all participating institutes recognize the validity of each other's calibration and measurement certificates for the quantities, ranges and measurement uncertainties specified in Appendix C (for details see http://www.bipm.org).	CIPM MRA
BAM holds an accreditation as a reference material producer according to ISO 17034. This accreditation is valid only for the scope as specified in the certificate D-RM-11075-01-00. DAkkS is a signatory of the multilateral agreement (MLA) between EA, ILAC and IAF for mutual acceptance.	DAkkS Deutsche Akkreditierungsstelle D-RM-11075-01-00

This Certified Reference Material is offered by:

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